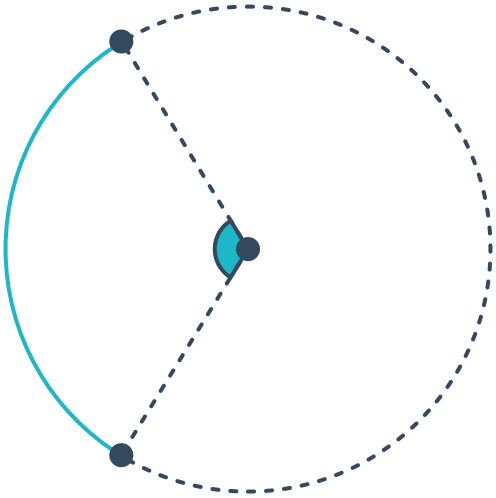
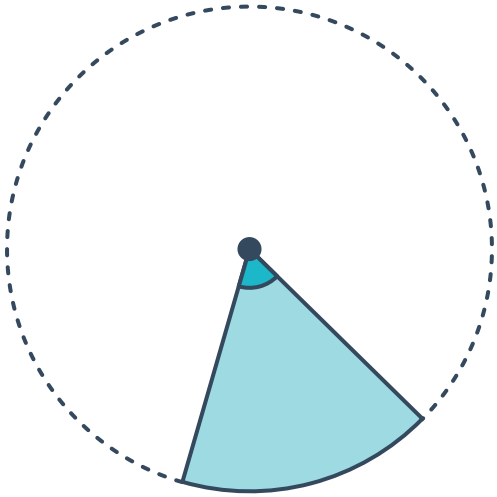
Arc length

1 Name the indicated parts of the following circles:

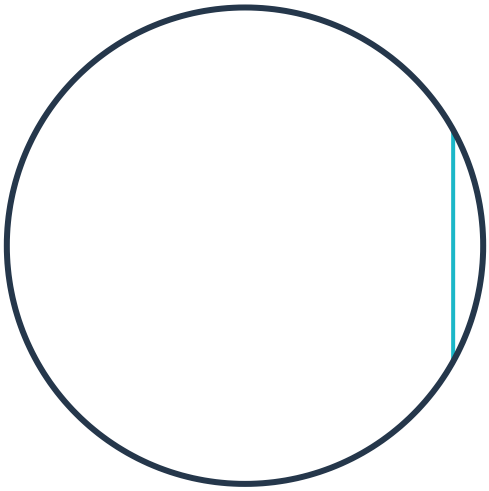
a



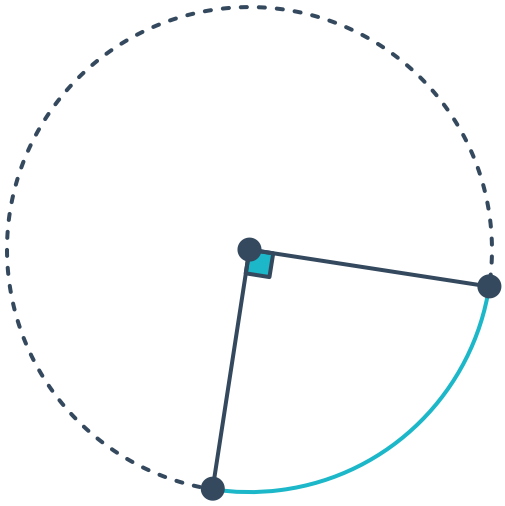
b



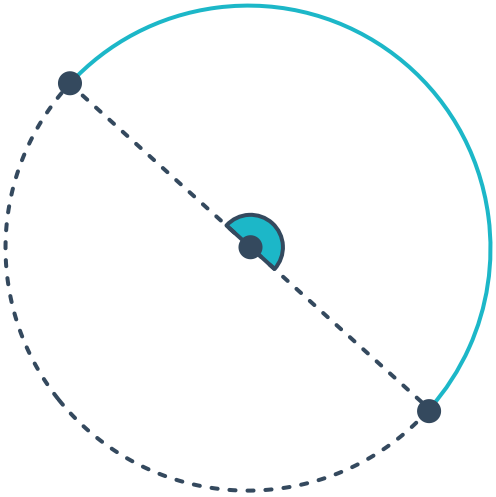
c



d



e



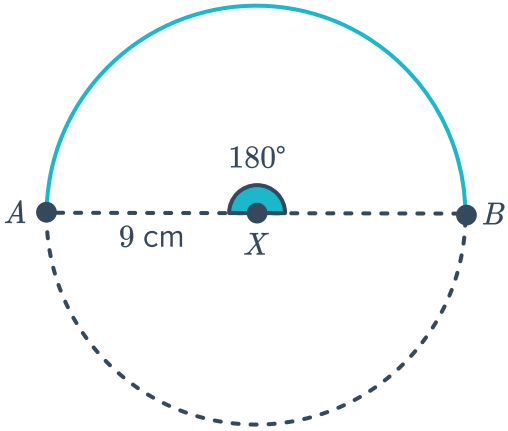
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2 For each of the following circles:

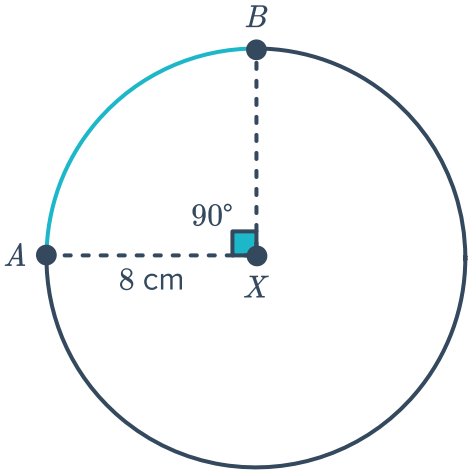
i Find the fraction of the whole circle that lies on the arc AB.

ii Find the exact length of the arc AB

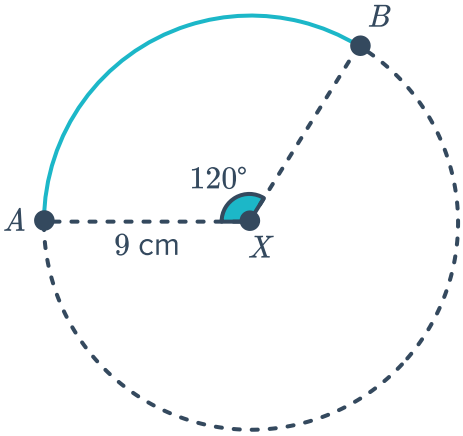
a



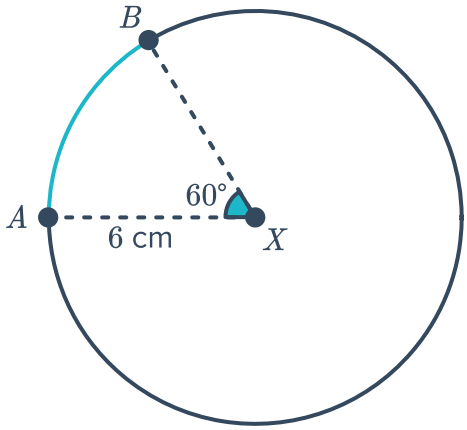
b



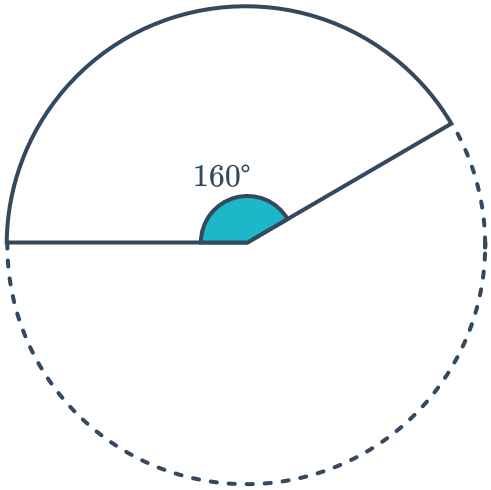
c



d

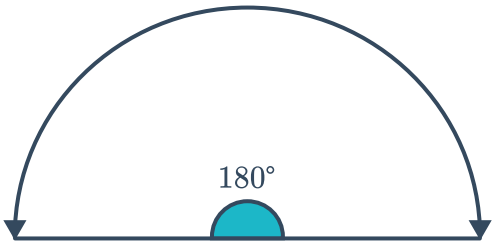


3 If the circumference of the circle is equal to 144 cm, find the length of the arc of the following sector:



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4 If the circumference of the full circle is equal to $40\ \text{cm}$, find the length of the arc of the following semicircle:

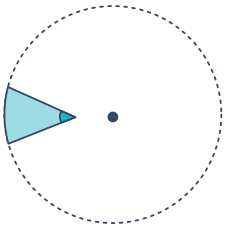


5 A sector of a circle with radius $29\ \text{m}$ is formed from an angle of size 131°. Find the exact length of the arc.

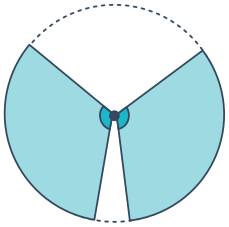
Area and perimeter of a sector

6 Determine whether the following diagrams represent a single sector:

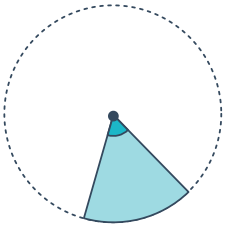
a



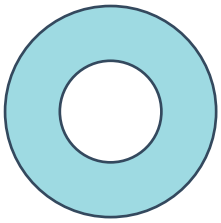
b



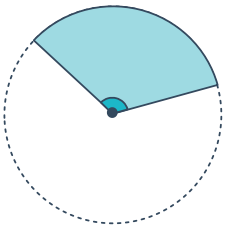
c



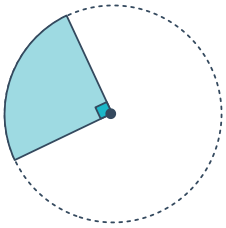
d



e



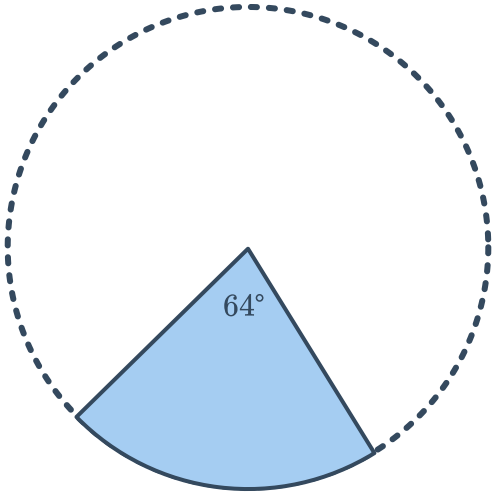
f



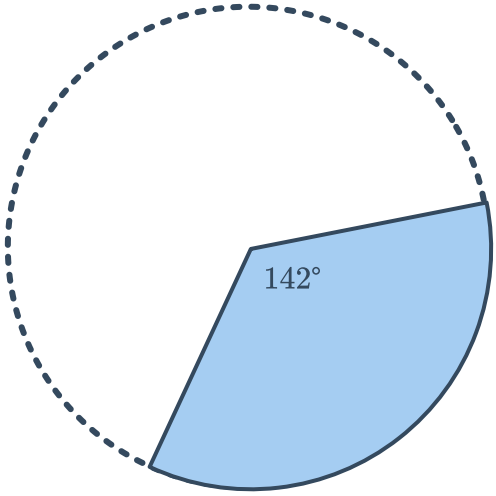
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7 Find the fraction of the circle's area covered by each sector:

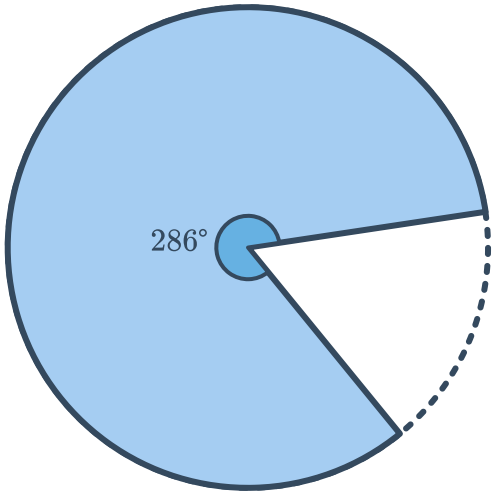
a



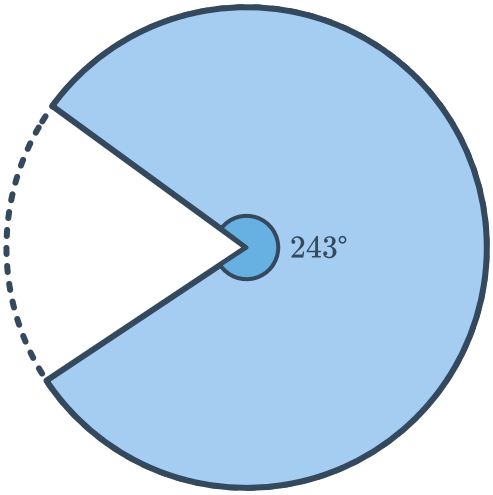
b



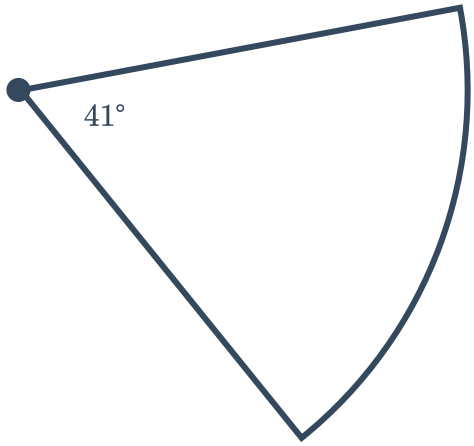
c



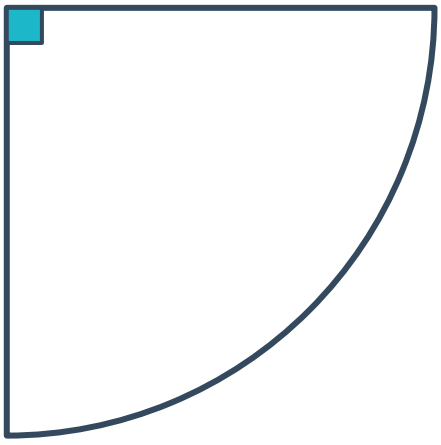
d



e



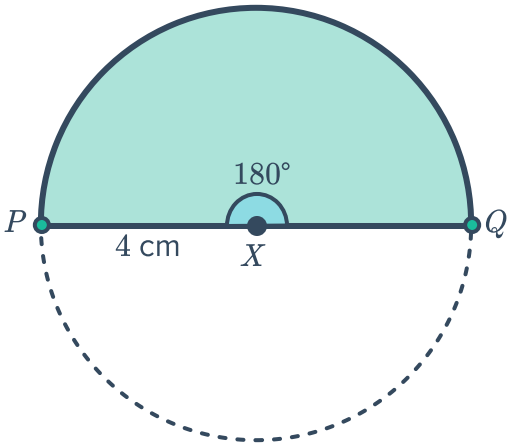
f



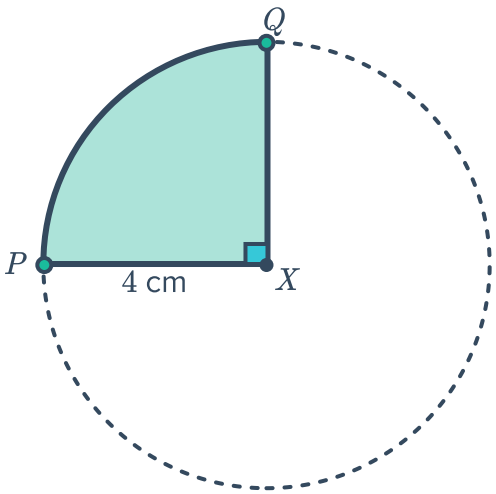
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8 Find the area of the following sectors to two decimal places:

a



b

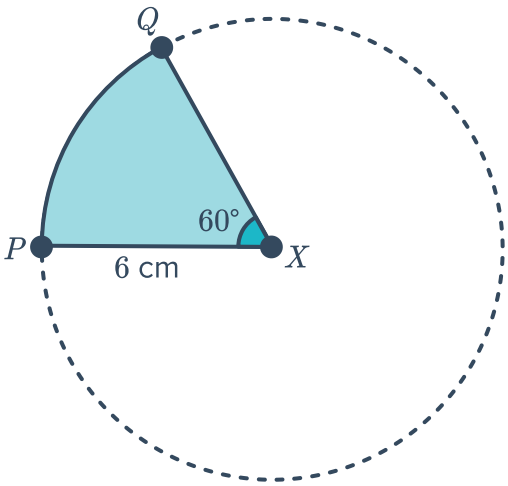


9 For each of the following circles:

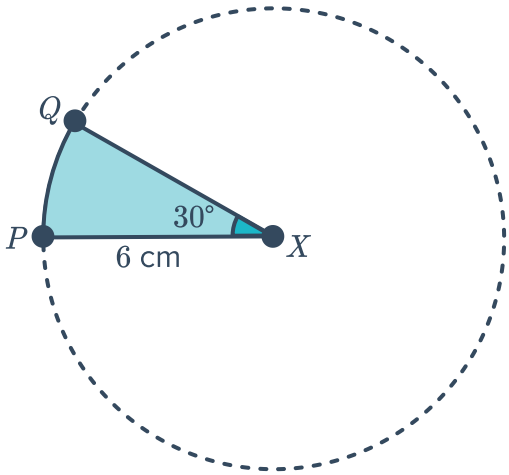
i Find the fraction of the circle's area covered by the sector.

ii Find the exact area of the sector.

a



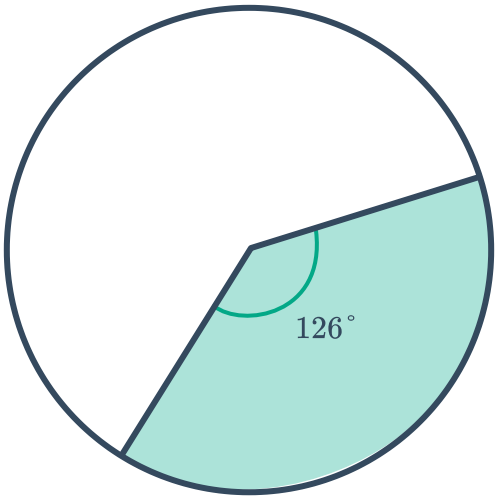
b



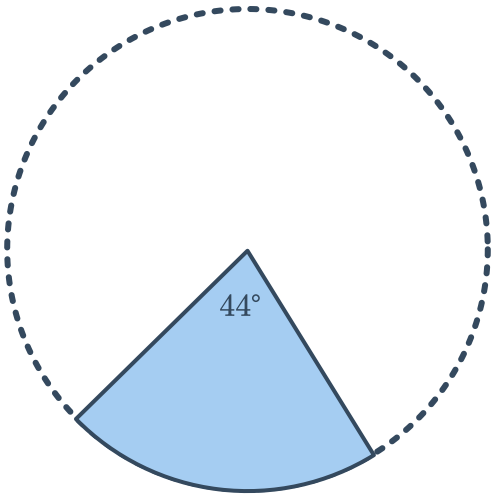
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10 Find the area of the shaded sectors, correct to two decimal places:

a Area of the circle is $10\ \text{cm}^2$.



b Area of the circle is $25\ \text{cm}^2$.



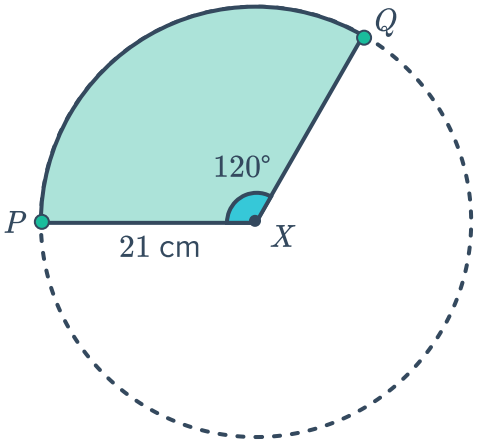
11 For each of the following circles:

i Find the fraction of the circle's area covered by the sector.

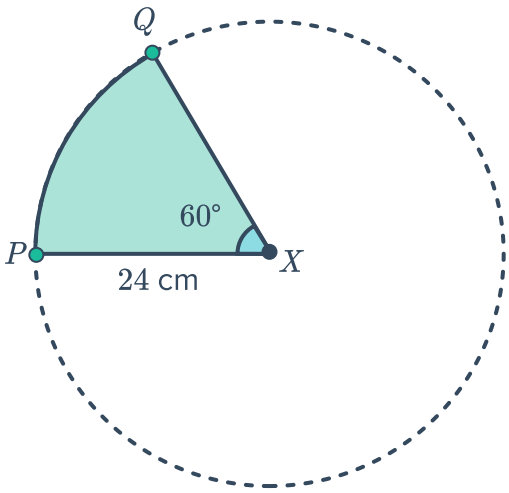
ii Find the perimeter of the sector to two decimal places.

iii Find the area of the sector to two decimal places.

a

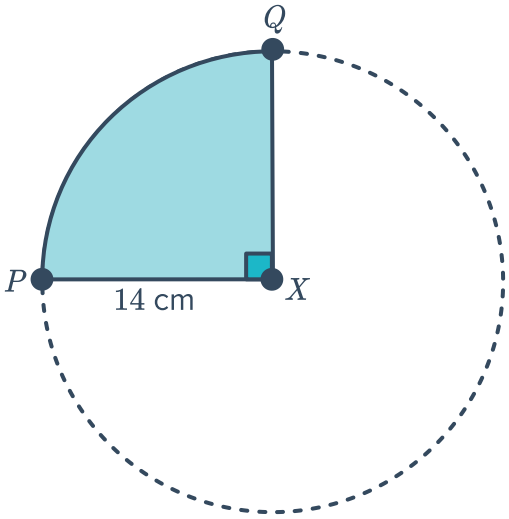


b



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12 The sector in the given diagram has an angle of 90° and a radius of 14 cm:

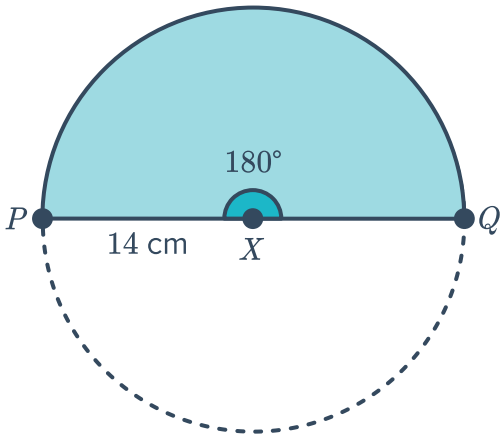


a Find the fraction of the circle's area covered by this sector.

b Find the exact length of the arc PQ.

c Find the perimeter of the sector, rounded to two decimal places.

13 The sector in the given diagram is a semicircle with a radius of 14 cm:



Find the exact perimeter of the sector.

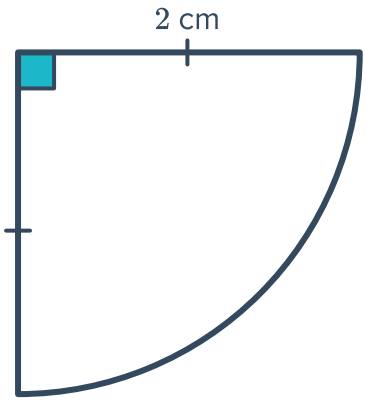
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14 For each of the following sectors:

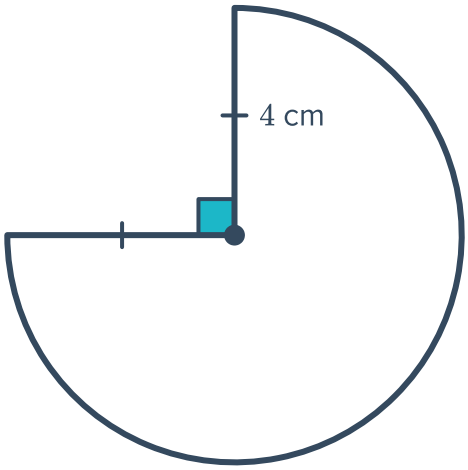
i Find the perimeter to the nearest whole number.

ii Find the area to the nearest whole number.

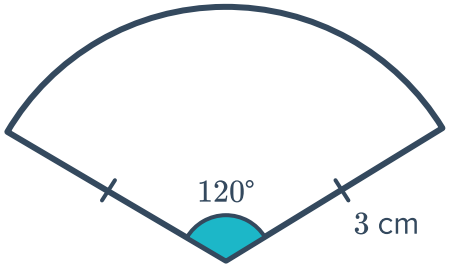
a



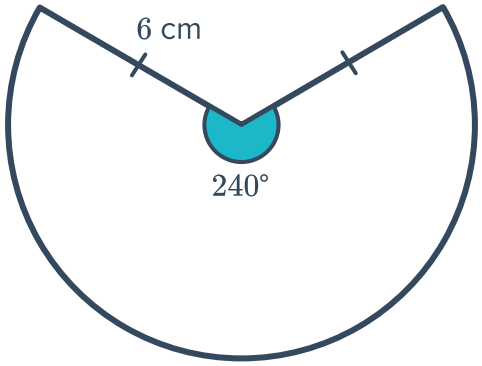
b



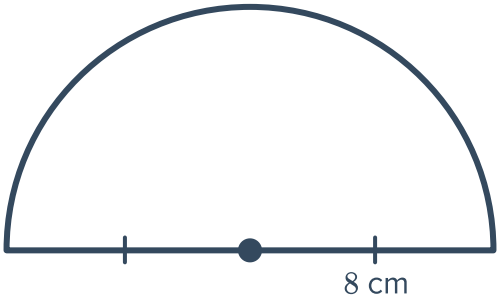
c



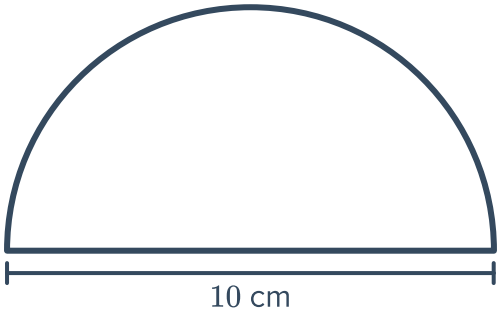
d



e



f



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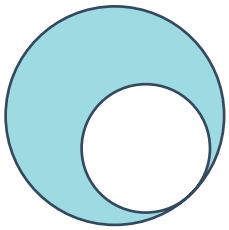
15 Find the perimeter of the following, correct to two decimal places:

- **a** A semicircle with radius 2 cm.
- **b** A semicircle with radius 19 cm.
- **c** A semicircle with diameter 8 cm
- **d** A semicircle with diameter 17.23 cm.
- **e** A sector equal to one quarter of a circle with radius 5 cm.
- **f** A sector equal to one quarter of a circle with radius 17 cm.
- **g** A sector equal to three quarters of a circle with radius 42 cm.
- **h** A sector equal to one third of a circle with radius 9 cm.
- **i** A sector equal to one third of a circle with radius 2.74 cm.
- **j** A sector equal to two thirds of a circle with radius 19 cm.

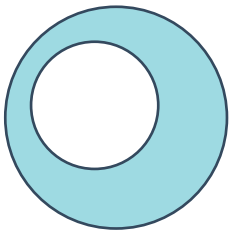
Annulus

16 Determine whether the following diagrams represent an annulus:

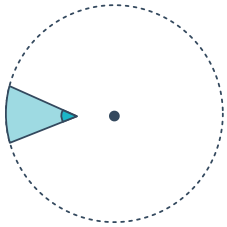
a



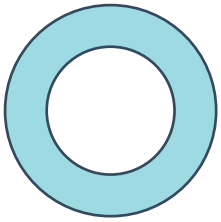
b



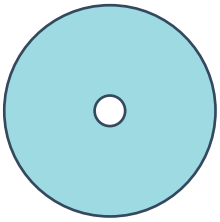
c



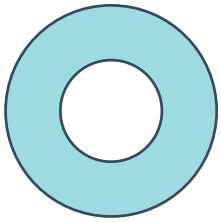
d



e



f



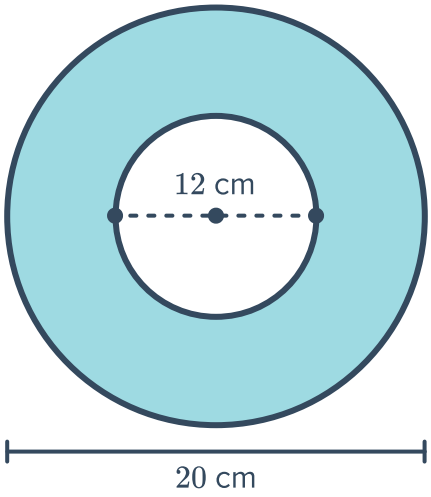
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17 For each annulus below:

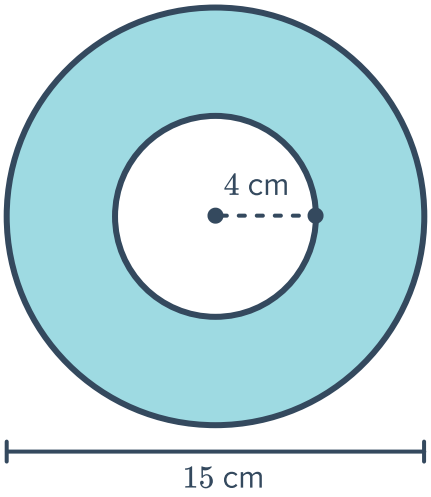
i Find the inner radius.

ii Find the outer radius.

a



b



18 Consider the following annulus:

a Find the inner radius.

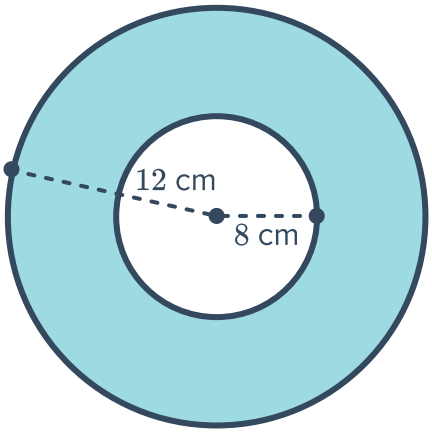
b Find the outer radius.

c Find the circumference of the outer circle, to one decimal place.

d Find the circumference of the inner circle, to one decimal place.

e Find the perimeter of the annulus, to one decimal place.

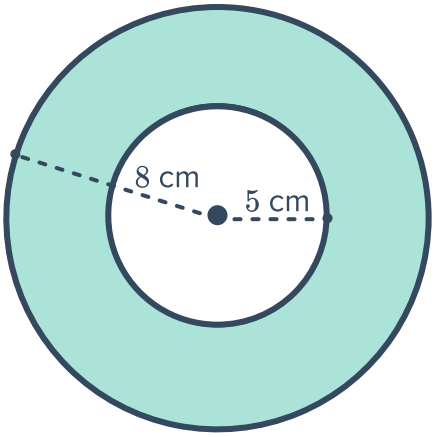
f Find the area of the annulus, to one decimal place.



19 Consider the following annulus:

a Find the exact area of the annulus.

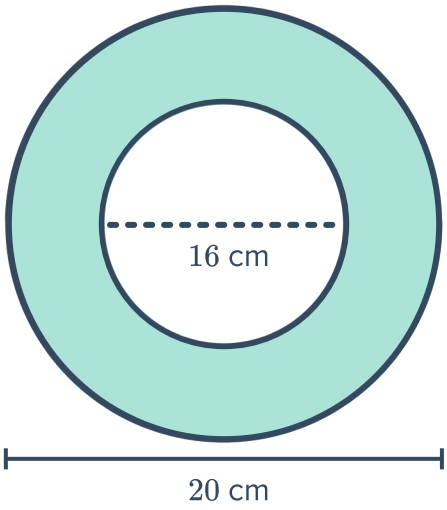
b Find the area of the annulus, rounded to two decimal places.



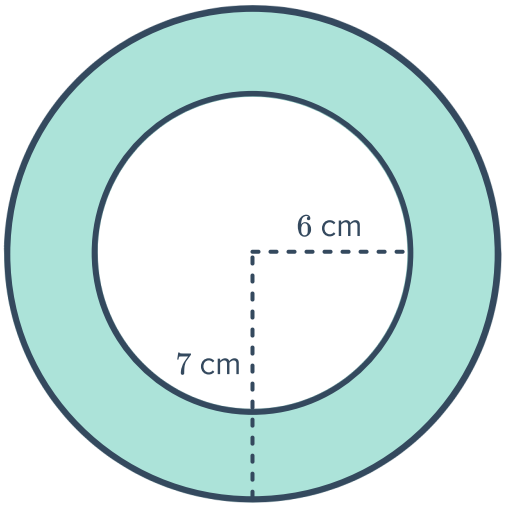
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20 Find the shaded area in the following figures, correct to one decimal place:

a

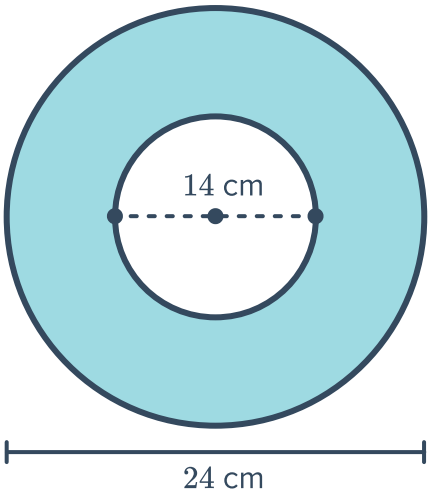


b

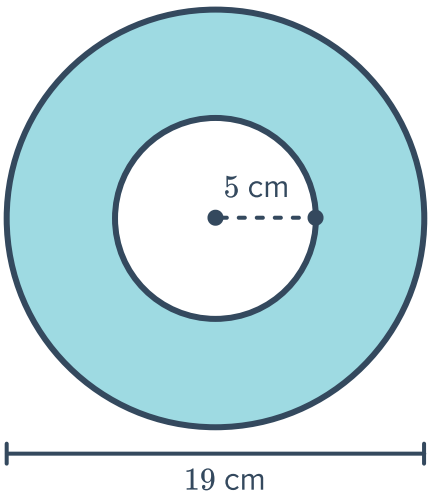


21 The following annulus has an inner diameter of 14 cm and an outer diameter of 24 cm:

Find its exact perimeter.



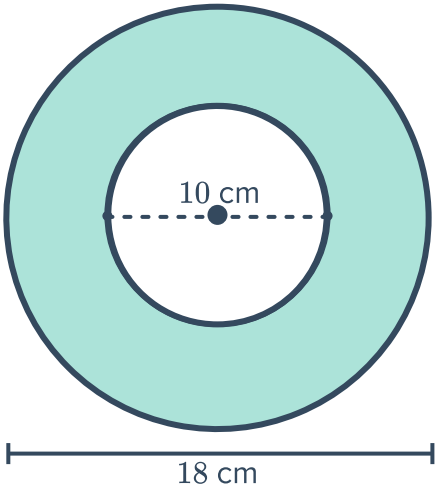
22 Find the perimeter of the following annulus, rounded to two decimal places:



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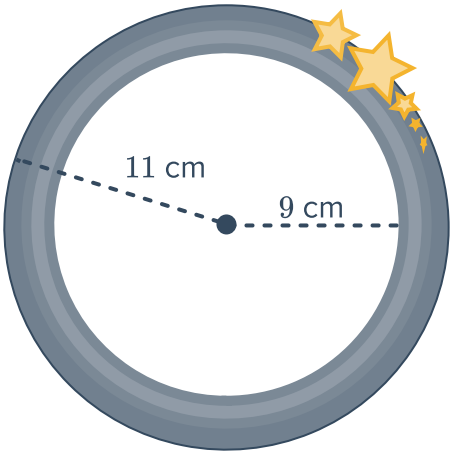
23 The following annnulus has an inner diameter of 10 cm and an outer diameter of 18 cm:

Find its exact area.



24 Tina, Brad, and Homer are designing a logo for their astronomy club.

Brad proposes that their logo should be in the shape of an annulus to represent an eclipse, as shown in the first diagram:



a Find the area of the annulus in the logo, correct to two decimal places.

Tina likes this idea but proposes that they shift the hole in the annulus to the side so that it better resembles an eclipse, as shown in the second diagram:



b Brad claims that this will change the area of the annulus. Is he correct? Explain your answer.

c Homer claims that the logo will no longer be an annulus. Is he correct? Explain your answer.

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